

## WATERWATCH · REQUIREMENTS

# WaterWatch requirements

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## DOCUMENT CONTROL

|             |                                         |         |                |
|-------------|-----------------------------------------|---------|----------------|
| Document ID | WWT-REQ-001                             | Version | 0.4            |
| Status      | <b>Draft</b>                            | Date    | 2026-09-25     |
| Author      | Amish Chadha                            | License | CERN-OHL-S-2.0 |
| Project TRL | <b>TRL 3</b> Proof of concept           |         |                |
| Repository  | github.com/BoujeeEnjinia1701/waterwatch |         |                |
| Source      | docs/03-requirements.md @ eddc63e       |         |                |

## REVISION HISTORY

| Version | Date       | Author       | Change                                                                                                          |
|---------|------------|--------------|-----------------------------------------------------------------------------------------------------------------|
| 0.4     | 2026-09-25 | Amish Chadha | Recommendations accepted by Amish (DDR-002)                                                                     |
| 0.3     | 2026-09-25 | Amish Chadha | For TRL 3, status of every requirement from WWT-CAL-001; decisions from WWT-DDR-001 recorded; targets unchanged |
| 0.2     | 2026-09-25 | Amish Chadha | First measurable requirements for TRL 2, with concept status                                                    |
| 0.1     | 2026-09-24 | Amish Chadha | Initial scaffold                                                                                                |

# WaterWatch requirements

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These requirements were first set for the concept and are checked by calculation in WWT-CAL-001 v0.2. Version 0.4 applies Amish's decisions of 2026-09-25 (WWT-DDR-002): R1 is relaxed, R11 is restated for a footing cast on the survey visit, R5 gains a 15 min confirming reading and R12 now states that the pH probe variant is priced per variant site. The targets are still not user-validated needs and will be revised after co-design. On paper, no requirement is missed outright; four are at risk (R1, R10, R2 and R9), six are met by calculation and three by design. In v0.3, R1 and R10 were not met and R11 was at risk. Amish's decisions are in WWT-DDR-001 and WWT-DDR-002.

*Table 1. Requirements and TRL 3 status (from WWT-CAL-001 v0.2, Table 4), at-risk items first.*

| ID  | Requirement                      | Target                                                                                                                                                                                                                                                                               | Verification                                                                                               | TRL 3 status                                                                                                                                                                                        |
|-----|----------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| R1  | Measure free chlorine at the tap | 0 to 2.0 mg/L; within $\pm 0.2$ mg/L or $\pm 25$ %, whichever is greater, at pH 6.5 to 8.5 and 5 to 40 °C (relaxed from $\pm 0.1$ mg/L below 1.0 mg/L and $\pm 15$ % above, WWT-DDR-002). Sites whose pH is above 7.5 or moves more than 0.2 between visits use the pH probe variant | Error budget (WWT-CAL-001, E); later side-by-side with a DPD comparator                                    | <b>At risk:</b> met in 8 of 8 cases at 25 °C before drift, with at least 0.13 mg/L left for drift; 1.5 mg/L at pH 7.5 and 40 °C misses by 0.006 mg/L; sensor drift unknown                          |
| R10 | Be maintainable by a caretaker   | One visit per month or less, 30 min or less, no special tools: clean the cell, compare with a DPD kit, adjust calibration from a phone or keypad                                                                                                                                     | Task analysis (WWT-CAL-001, J)                                                                             | <b>At risk:</b> the visit takes 30 min, at the limit, and a monthly interval needs chlorine drift below about 0.13 mg/L per month under the relaxed R1 (was 0.04 mg/L), which no verified data show |
| R2  | Measure turbidity                | 0 to 100 NTU, reporting "above 100 NTU" beyond; within $\pm 0.5$ NTU or $\pm 10$ %, whichever is greater                                                                                                                                                                             | Published open-source design; bubble and settling calculation (WWT-CAL-001, F); later stabilized standards | <b>At risk:</b> the 30 s wait clears 100 $\mu$ m bubbles at 5 °C; window fouling between cleans is unknown                                                                                          |
| R9  | Survive outdoors                 | Electronics IP65 or better; ambient 0 to 45 °C in direct sun; UV-stable parts; flow cell opaque to daylight; battery charging blocked below 0 °C and above 45 °C                                                                                                                     | Thermal estimate (WWT-CAL-001, C); parts selection                                                         | <b>At risk:</b> with the sun shield the inside peaks at 50.8 °C on a 45 °C day (69.8 °C without); the shield factor is assumed                                                                      |
| R5  | Alert the people who act         | SMS to up to three numbers within 15 min of a confirmed crossing (two consecutive readings, the second taken 15 min after the first crossing, WWT-DDR-002): free chlorine below 0.2 mg/L, turbidity above 5 NTU, device fault or low battery; thresholds adjustable per site         | Latency calculation (WWT-CAL-001, H)                                                                       | Met on paper where there is coverage: 13.0 min worst case with three tries at 3 min spacing; 77 min from the event (was 122 min)                                                                    |
| R6  | Report and keep data             | Upload readings at least every 4 h over LTE-M, NB-IoT or 2G (LoRaWAN variant); keep 90 days on board if the link fails; open CSV or JSON format; works without a cloud service                                                                                                       | Data budget (WWT-CAL-001, G)                                                                               | Met on paper: 0.86 MB per month; 90 days in 138 kB                                                                                                                                                  |
| R7  | Run on sunlight alone            | 7 days or more with no sun from full; back to full in 3 clear days or fewer                                                                                                                                                                                                          | Energy budget (WWT-CAL-001, A and B)                                                                       | Met on paper: 28.6 days (24.3 at 0 °C); 1.18 clear days to refill; hot days rely on the sun shield (R9)                                                                                             |
| R8  | Use little treated water         | 20 L/day or less at hourly sampling                                                                                                                                                                                                                                                  | Flow calculation (WWT-CAL-001, D)                                                                          | Met on paper, thin margin: 18.0 L/day nominal, 19.8 L at the regulator's tolerance                                                                                                                  |
| R11 | Install simply and safely        | Two people, 2 h or less on installation day, with the pole footing dug and cast on                                                                                                                                                                                                   |                                                                                                            | Met on paper: 65 min critical path on installation day (140                                                                                                                                         |

| ID  | Requirement                       | Target                                                                                                                                                    | Verification                                         | TRL 3 status                                                                              |
|-----|-----------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------|-------------------------------------------------------------------------------------------|
|     |                                   | the survey visit (WWT-DDR-002); hand tools only; one tee and isolation valve on the riser; backflow prevented; wetted parts food-safe                     | Task analysis (WWT-CAL-001, J); design review        | min if done in one visit); the design parts are met                                       |
| R12 | Stay within the concept budget    | Parts \$300 or less per base unit, excluding the shared calibration kit, airtime and the pH probe variant, which is priced per variant site (WWT-DDR-002) | Priced BOM (WWT-CAL-001, K)                          | Met on paper: \$283 per base unit; \$343 to \$383 at a pH probe variant site              |
| R3  | Measure water temperature         | 0 to 50 °C, within $\pm 0.5$ °C                                                                                                                           | Sensor datasheet                                     | Met by design                                                                             |
| R4  | Sample on a schedule              | One reading per hour by default; configurable from 15 min to 24 h                                                                                         | Energy at the 15 min schedule (WWT-CAL-001, A and B) | Met by design (15 min schedule: 1.589 Wh/day, 9.7 days on battery, but 72 L/day of water) |
| R13 | Report measurements, not verdicts | Messages state what was measured and when; the device never says the water is "safe"                                                                      | Design review of alert wording                       | Met by design                                                                             |

## ■ Assumptions

- Site: public tapstand on a piped, chlorinated rural scheme, with a supply pressure of 0.5 to 4 bar (7 to 58 psi) at the riser. Tapstands come first; tank outlets and kiosks later (WWT-DDR-001, D8).
- Alert thresholds follow the field guidance summarized in WWT-PRB-001: at least 0.2 mg/L free chlorine at the point of delivery, and chlorination losing effectiveness as turbidity rises. The defaults of 0.2 mg/L and 5 NTU were decided by Amish on 2026-09-25 (WWT-DDR-001, D6) and remain adjustable per site.
- The caretaker's DPD test is the field reference for free chlorine. The v0.3 target of  $\pm 0.1$  mg/L matched the resolution of common DPD comparator kits, which WWT-CAL-001 showed made it as tight as the reference itself (the comparator alone takes  $\pm 0.07$  mg/L); R1 was relaxed to  $\pm 0.2$  mg/L or  $\pm 25$  % for that reason (WWT-DDR-002).
- pH affects the split between hypochlorous acid and hypochlorite, and amperometric sensors respond mainly to hypochlorous acid. R1 is assessed with a site pH entered at installation and checked monthly (WWT-DDR-001, D2), assumed to move no more than  $\pm 0.2$  between visits. Under WWT-DDR-002 the pH probe variant (read to  $\pm 0.1$ ) is fitted at sites whose pH is above 7.5 or moves more than 0.2 between visits; the cell lid carries a plugged spare port for it.
- Budget: the \$300 in `project.yaml` covers one base unit's parts. A shared calibration kit (about \$60), airtime (about \$1 to \$3 per month) and, at variant sites only, the pH probe and interface (about \$60 to \$100) are listed separately; all are estimates.
- Installation: the pole footing is dug and cast on the survey visit, so the pole is set and cured before installation day (WWT-DDR-002).

**Safety:** Meeting these requirements does not make the water safe to drink or the unit safe to install. The sample line connects to a pressurized drinking water supply and the unit contains a lithium iron phosphate battery; see WWT-PRC-001, Safety.